

01)

Images

Graphics

- | | |
|---|---|
| * Captured from real world situation using cameras, scanners or sensors | * Created digitally using mathematical formulas, vector software or code. |
| * Lose quality and become pixelated when enlarged. | * Infinitely scalable without any loss of resolution. |
| * Generally larger | * Generally smaller |

02)

1D

2D

3D

- | | | |
|---------------------------------------|------------------------------------|---|
| * Represents data along a single axis | * Represents data across two axes | * Represents data across three axes |
| * Only measure length or distance | * Measure length and width | * Measure length, width and depth |
| * Exist as a straight line. | * Objects on the pictures are flat | * Objects can be rotated, navigated and viewed from any angle |

03)

$$a. \text{Number of pixels} = 480 \times 320 = 153\ 600$$

$$\text{Required bits} = 153\ 600 \text{ bits}$$

$$= \frac{153\ 600}{8} = 19\ 200 \text{ Bytes} = \frac{19\ 200}{1024} = 18.75 \text{ KB}$$

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b. Number of pixels = $512 \times 1024 = 524\,288$

Required bits = $524\,288 \times 8 = 4\,194\,304$ bits

$$= \frac{4\,194\,304}{8}$$

$$= 524\,288 \text{ Bytes}$$

$$= \frac{524\,288}{1024} = 512 \text{ KB}$$

$$= \frac{512}{1024} = 0.5 \text{ MB} //$$

c. Number of pixels = $2048 \times 1024 = 2\,097\,152$

Required bits = $2\,097\,152 \times 8 = 16\,777\,216$ bits

$$= \frac{16\,777\,216}{8} = 2\,097\,152 \text{ Bytes}$$

$$= \frac{2\,097\,152}{1024} = 2\,048 \text{ KB}$$

$$= 2 \text{ MB} //$$

$$d. \text{ Number of pixels} = 2048 \times 1024 = 2\,097\,152$$

$$\text{Required bits} = 2\,097\,152 \times 8 = 16\,777\,216 \text{ bits}$$

$$= \frac{16\,777\,216}{8} = 2\,097\,152 \text{ Bytes}$$

$$= \frac{2\,097\,152}{1024} = 2048 \text{ KB}$$

$$= \frac{2048}{1024} = 2 \text{ MB} //$$

4) YUV color model

- A color encoding system that separates visual information into brightness (luminance) and color (chrominance).

- Y (Luminance)

- Stores brightness and grayscale detail

- U (chrominance)

- stores color difference for blue

- V (chrominance)

- stores color difference for red.

5) YIQ color model

- It separate visual data into brightness (luminance) and color (chrominance) components.

• Y (Luminance)

- store brightness and grayscale detail

• I (In-phase)

- Store color along the orange-cyan axis

• Q (Quadrature)

- store color along the magenta-green axis

6) YCbCr color model

- a family of color spaces used in digital video and digital photography systems to separate brightness from color information.

• Y (Luminance)

- Represent the brightness or grayscale information of the image.

• C_b (Blue-difference Chrominance)

- Represent the difference between the blue component and luminance $(B - Y)$

• C_r (Red-difference Chrominance)

- Represent the difference between the red component and luminance $(R - Y)$

7) Morphing

- A digital visual effect that smoothly transforms one image or 3D model into another through a seamless transition.

8) Dithering

- A computer graphics technique used to create the illusion of smooth color gradients or additional color shades when displaying images on hardware with a limited color palette.

9) Lossy compression

Lossless compression

* Permanently deletes less noticeable data to maximize file size reduction

* Retains all original data without discarding any information.

* The original file cannot be restored bit for bit.

* Decompresses into an exact copy of the original file.

* Results in high compression rates with a slight drop in quality.

* Results in ^{lower} compression rates with zero loss in quality.

eg:- JPEG, MP3, MP4, JPG, GIF.

eg:- PNG, ZIP, RAW, BMP.

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10)

1, 7, 2

1, 1, 5, 1, 2

1, 1, 5, 1, 2

1, 1, 5, 1, 2

1, 1, 5, 1, 2

1, 7, 2

3, 3, 4

0, 9, 1

0, 9, 1

11)



12)

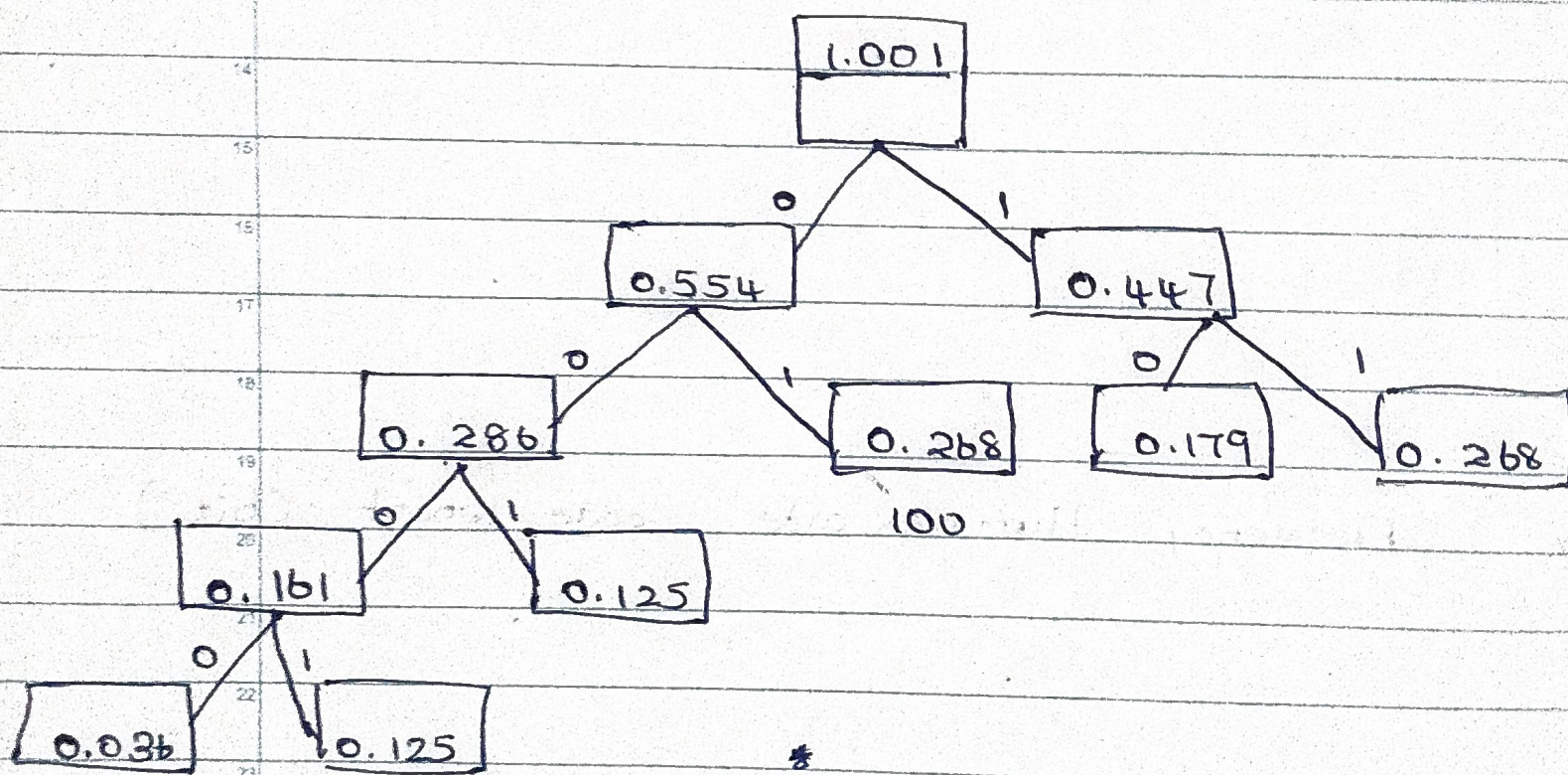
color

frequency

Probability

90	7	$\frac{7}{56} = 0.125$
98	15	$\frac{15}{56} = 0.268$
100	15	$\frac{15}{56} = 0.268$
102	2	$\frac{2}{56} = 0.036$
105	10	$\frac{10}{56} = 0.179$
112	7	$\frac{7}{56} = 0.125$

102	112	90	105	98	100
0.036	0.125	0.125	0.179	0.268	0.268



90	001
98	11
100	01
102	0000
105	10
112	0001

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01000100100110110111

101101111101011

01000100100110110111

00000000010010001011001

1101010010010011111

101101111101011

010100010001000101010001

Image size in bits = $20 + 16 + 20 + 22 + 19 + 16 + 24 = 137$

Huffman table size in bits = $(8 \times 6) + 3 + 2 + 2 + 4 + 2 + 4$
 $= 65$

original image size in bits = $56 \times 8 = 448 \text{ bit //}$

13)	Row 1	Row 2	Row 3	Row 4
	10	010	0111	0111
	11	010	0111	0111
	0110	10	10	10
	00	10	11	11
	00	11	11	11
	10	11	11	11
	10	11	11	11
	11	00	10	10
	11	00	00	0110
	10	0110	00	0110
	10		010	010
				00
				00
				11

$$\text{compressed image size} = 24 + 24 + 27 + 37$$

$$= 112 \text{ bits}$$

$$\text{Huffman Table size} = (6 \times 8) + (3 + 2 + 4 + 2 + 4 + 2)$$

$$= 48 + 17$$

$$= 65 \text{ bits}$$

$$\text{Total required bits to store image}$$

$$= 112 + 65$$

$$= 117 \text{ bits} //$$